

The Transmission Control Protocol (TCP) is a core protocol of the Internet Protocol Suite and operates at the transport layer. TCP provides reliable, ordered, and error-checked delivery of data between applications running on hosts communicating over an IP network. It uses a three-way handshake process to establish a connection before data transmission begins. Flow control in TCP is managed using a sliding window mechanism, which ensures that the sender does not overwhelm the receiver. Additionally, TCP implements congestion control algorithms such as slow start, congestion avoidance, fast retransmit, and fast recovery to maintain network stability. Due to these features, TCP is widely used in applications where data integrity and reliability are critical, such as web browsing and file transfer.